

Mustakim Shikalgar

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EDUCATION

Arizona State University

Master of Science in Software Engineering; GPA: 3.75/4.0

Tempe, AZ

May 2026

TECHNICAL SKILLS

Languages: Python, Java, TypeScript/JavaScript, C++, SQL

Backend & Systems: FastAPI, Django, REST APIs, gRPC, microservices, distributed systems, circuit breakers, async programming

AI & ML: LangGraph, LangChain, LangSmith, RAG, prompt engineering, multi-agent systems, PyTorch, scikit-learn

Infrastructure: Kubernetes, Docker, Prometheus, OpenTelemetry, Redis, NATS, Nginx, CI/CD, Grafana

Cloud & Data: Google Cloud Platform, AWS, PostgreSQL, MongoDB, pgvector, Neo4j, Kafka

PROFESSIONAL EXPERIENCE

Software Engineer (Technical Architecture)

Jan 2026 – May 2026

METY Legal AI Platform (Industry Capstone), MyEdmaster

Arizona State University

- Owned technical architecture for a **6-person team** across 7 Agile sprints; delivered a fully functional AI-powered legal platform covering **20+ practice areas** with zero critical failures during testing and validation
- Built **REST API layer** for authentication, session management, document ingestion, and knowledge retrieval; applied repository and strategy design patterns with unit and integration test coverage across all core flows
- Engineered **RAG architecture** with multi-turn memory threading and hallucination mitigation via self-assessment scoring; reduced cost-per-message by **82% to \$0.0044/query** through targeted model routing between GPT-4o and GPT-4o-mini, validated end-to-end with LangSmith
- Deployed **containerized microservices** on **Google Cloud Platform** with Kubernetes orchestration, Django behind Nginx for load balancing, and stateless FastAPI AI layer for horizontal scale

PROJECTS

AegisFlow - LLM Reliability Platform | Python, FastAPI, Kubernetes, Redis, NATS

Apr 2026 – Present

- Architected **8-microservice reliability middleware** (9,500+ lines) wrapping LLM providers with circuit breakers, confidence scoring, output repair; deployed on **Kubernetes** with HPA, PodDisruptionBudgets, 5-replica production configuration
- Engineered **5-component confidence model** (structural parse + Jaccard grounding + LLM critique + Bayesian-shrunk historical rates + anomaly penalty) for deterministic routing over non-deterministic AI outputs
- Implemented **sliding-window circuit breaker** with exponential cooldown (15s to 120s); built **atomic Redis rate limiting** via EVALSHA; deployed **11 Prometheus metrics**, OpenTelemetry tracing, and Grafana dashboards

Distributed Key-Value Store | Java, Raft Consensus, gRPC, Protocol Buffers, Docker

Jan 2025 – Dec 2025

- Implemented fault-tolerant distributed database using **Raft consensus** across **5-node cluster** with leader election, log replication, linearizable consistency; applied interface-driven OOP design
- Built gRPC service layer with automatic leader discovery, client-side caching; achieved **sub-10ms p99 latency** through lease-based optimization and tunable consistency (CP vs AP trade-offs)

Missing Persons Investigation Platform | React, FastAPI, PostgreSQL, RDFLib, SPARQL

Aug 2024 – Dec 2024

- Architected semantic knowledge graph integrating **10,000+ NamUs records** with RDF ontology; published at **IEEE COMPSAC 2025** (27% acceptance) demonstrating **3x improvement** in case correlation over relational schemas
- Replaced managed GraphDB (**\$50/mo to \$0**) with FastAPI and in-memory RDFLib; achieved **sub-100ms queries**, built React frontend with full endpoint test coverage

ACHIEVEMENTS

LeetCode: 860+ problems (Top 15%), 130+ Hard | **Publication:** IEEE COMPSAC 2025 (27% acceptance) |
Mathematics Olympiad: Silver Medalist